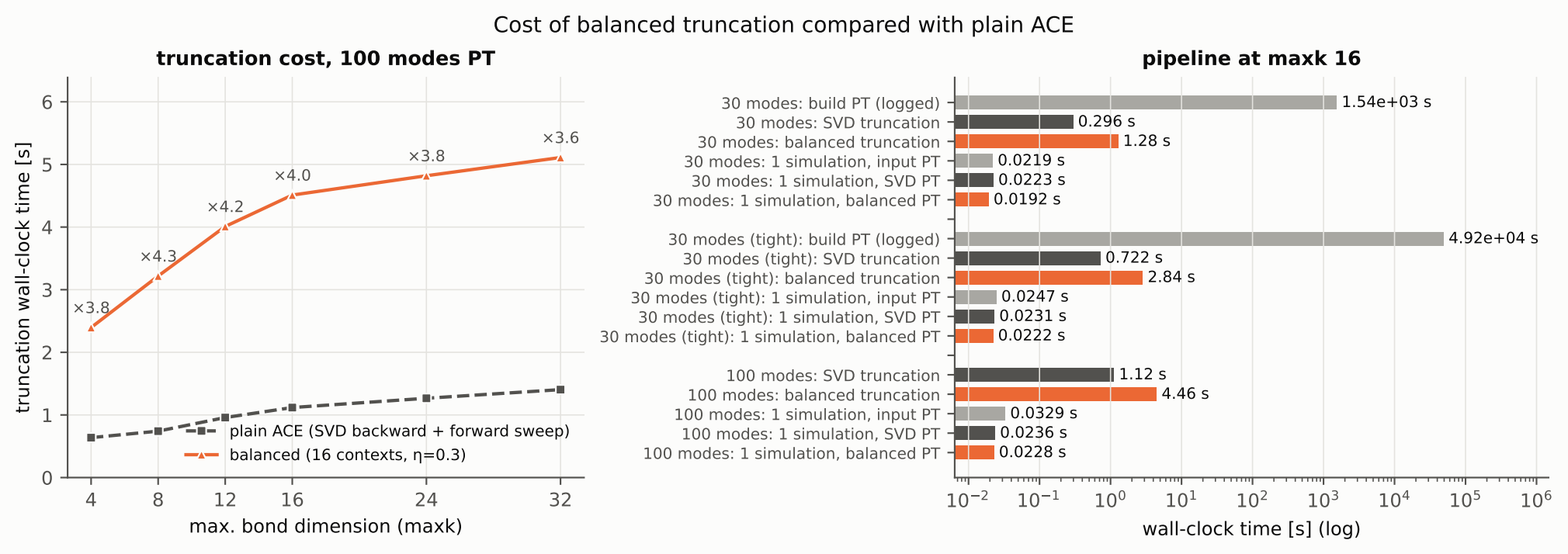
Cost of balanced truncation compared with plain ACE
truncation cost, 100 modes PT
truncation wall-clock time [s]
max. bond dimension (maxk)
plain ACE (SVD backward + forward sweep)
balanced (16 contexts, η=0.3)
×3.8
×4.3
×4.2
×4.0
×3.8
×3.6
pipeline at maxk 16
30 modes: build PT (logged)
30 modes: SVD truncation
30 modes: balanced truncation
30 modes: 1 simulation, input PT
30 modes: 1 simulation, SVD PT
30 modes: 1 simulation, balanced PT
30 modes (tight): build PT (logged)
30 modes (tight): SVD truncation
30 modes (tight): balanced truncation
30 modes (tight): 1 simulation, input PT
30 modes (tight): 1 simulation, SVD PT
30 modes (tight): 1 simulation, balanced PT
100 modes: SVD truncation
100 modes: balanced truncation
100 modes: 1 simulation, input PT
100 modes: 1 simulation, SVD PT
100 modes: 1 simulation, balanced PT
1.54e+03 s
0.296 s
1.28 s
0.0219 s
0.0223 s
0.0192 s
4.92e+04 s
0.722 s
2.84 s
0.0247 s
0.0231 s
0.0222 s
1.12 s
4.46 s
0.0329 s
0.0236 s
0.0228 s
wall-clock time [s] (log)